

Robust Precoding for Cooperative Massive MIMO LEO Satellite-Terrestrial Networks

Qingfu Kong , Li You , Huibin Zhou , Yongxiang Zhu ,
Yuanshuo Wang, and Xiqi Gao , *Fellow, IEEE*

Abstract—Low earth orbit (LEO) satellites, with advantages such as wide coverage and resilience to obstacles, make the establishment of a cooperative satellite-terrestrial network (STN) a feasible solution to address the limitations of terrestrial networks. In the cooperative STN, employing massive multiple-input multiple-output (MIMO) technology can fulfill the requirements for wide coverage and high-quality service. However, due to angle estimation errors in LEO satellite communication, acquiring channel state information (CSI) poses significant challenges. To tackle this issue, this paper explores robust precoding in the cooperative massive MIMO LEO STN. Specifically, we first introduce the imperfect CSI model. Then, we formulate the problem of maximizing the sum rate and propose a robust precoding scheme that designs the precoders for both base stations (BSs) and the LEO satellite, leveraging imperfect CSI and error distribution information. Simulation results demonstrate the validity of our robust precoding scheme.

Index Terms—Massive MIMO, LEO STN, imperfect CSI, robust precoding.

I. INTRODUCTION

As device connectivity and data traffic continue to grow, the development of sixth-generation (6G) wireless networks has become inevitable. Beyond achieving higher communication capacity, 6G networks should ensure ubiquitous coverage [1], [2]. Traditional terrestrial networks are primarily deployed in densely populated and developed regions, covering only about 6% of earth's surface. Consequently, users in remote areas often lack reliable communication services [3]. In contrast, satellite networks, with their inherent deployment flexibility, can effectively complement terrestrial networks, offering extensive coverage as well as enhanced data rates [4], [5].

Low earth orbit (LEO) satellites have attracted significant interest from both academia and industry due to their reduced latency and lower path loss compared to geosynchronous earth orbit (GEO) and medium earth orbit (MEO) satellites. In addition, massive multiple-input multiple-output (MIMO) technology, widely adopted in terrestrial networks, has shown great potential in enhancing spectral efficiency, energy efficiency and interference resistance when extended to LEO satellite networks [6]. The effectiveness of downlink precoding in LEO satellite systems is strongly influenced by the accuracy of channel state information (CSI) based on the angle of departure (AoD). However,

achieving perfect CSI in LEO satellite communication systems is challenging due to practical constraints, such as high mobility and significant propagation delays [5], [7], [8].

In recent years, integrated satellite-terrestrial networks (STNs), which leverage the strengths of both terrestrial and satellite networks, have gained significant research interest. Some studies have utilized cognitive radio (CR) techniques to manage satellite and terrestrial networks, treating them as primary and secondary networks, respectively, while focusing on minimizing interference from the secondary network to the primary one [9]. Other works have categorized users into terrestrial users and satellite users, served by base stations (BSs) and satellites, respectively [10]. However, these studies do not consider cooperative services involving both BSs and satellites for all users, not fully realizing the potential of the STN. Cooperative multicast methods between satellites and BSs have been explored [11], but they overlook the impact of channel outage and imperfect CSI.

The implementation of robust precoding design in STNs necessitates the modeling of imperfect CSI. Some studies have explored imperfect CSI modeling and robust precoding design in STNs. Specifically, Zhao et al. proposed an angular discretization method for robust precoding by modeling AoD uncertainties in downlink channels under known angular uncertainty regions [12]. In addition, Chu et al. developed a Bernstein-type inequality-based beamforming algorithm that handles satellite channel phase errors and BS channel errors using second-order Taylor approximations, assuming statistical error knowledge [13]. For channels with bounded channel errors, Zhang et al. transformed the parameter perturbations from channel estimation errors into deterministic forms through worst-case optimization and the S-procedure [14]. To the best of our knowledge, no existing studies have investigated robust precoding design in scenarios where both the LEO satellite and terrestrial BSs cooperatively serve the same users.

This paper investigates cooperative transmission for the massive MIMO LEO STN. We first model the AoD error of the LEO satellite downlink channel and the channel state on BSs' downlink channels. To mitigate potential signal degradation in terrestrial downlink channels, we propose a cooperative transmission framework where the LEO satellites and BSs jointly serve users. Under this framework, we derive the ergodic achievable rate for users. Subsequently, we formulate a sum-rate maximization problem and develop a robust precoding algorithm. Finally, extensive numerical simulations validate the effectiveness of the proposed robust precoding scheme, demonstrating its superiority over traditional robust baselines in practical LEO STN deployments.

II. SYSTEM MODEL

We consider a cooperative massive MIMO LEO STN model with a unicast communication architecture. In this network, a LEO satellite cooperates with N_B BSs to jointly serve K_u terrestrial user terminals (UTs) within the same frequency band. All users are located within the satellite's coverage area and can be simultaneously served by multiple BSs. A gateway (GW) manages the entire cooperative STN.

A. LEO Satellite Channel Model

The LEO satellite is outfitted with a uniform planar array (UPA) comprising $M_t = M_1 M_2$ antennas, where M_1 and M_2 represent the number of antennas evenly distributed along the x and y -axes, respectively. Based on a ray-tracing system model, the downlink spatial domain channel response of the LEO satellite to the j -th user in the

Received 4 January 2025; revised 3 May 2025 and 14 July 2025; accepted 19 August 2025. Date of publication 28 August 2025; date of current version 13 February 2026. This work was supported in part by the National Natural Science Foundation of China for Outstanding Young Scholars under Grant 62322104, in part by the Jiangsu Province Major Science and Technology Project under Grant BG2024005, in part by the Natural Science Foundation of Jiangsu Province under Grant BK20231415, in part by the Fundamental Research Funds for the Central Universities under Grant 2242022k60007, and in part by the Civil Aerospace Technology Pre-research Project under Grant D030301. The review of this article was coordinated by Dr. Boya Di. (*Corresponding author: Li You.*)

The authors are with the National Mobile Communications Research Laboratory, Southeast University, Nanjing 210096, China, and also with the Purple Mountain Laboratories, Nanjing 211100, China (e-mail: qfkong@seu.edu.cn; lyou@seu.edu.cn; zhouhb@seu.edu.cn; zhuix@seu.edu.cn; yswang@seu.edu.cn; xqgao@seu.edu.cn).

Digital Object Identifier 10.1109/TVT.2025.3603905

complex baseband at time t and frequency f is [5]

$$\mathbf{g}_j = \exp \{j2\pi [tv_j^{\text{sat}} - f\tau_j^{\text{min}}]\} \cdot g_j(t, f) \cdot \mathbf{v}_j, \quad (1)$$

$$g_j(t, f) = \sum_{l=0}^{L_{s,j}-1} g_{j,l} \exp \{j2\pi [tv_{j,l}^{\text{ut}} - f\tau_{j,l}^{\text{ut}}]\}, \quad (2)$$

where v_j^{sat} denotes the Doppler shift due to the satellite's mobility. $\tau_j^{\text{min}} = \min_l \{\tau_{j,l}\}$ denotes the minimum path delay, where $\tau_{j,l}$ represents the delay of the l -th path from the LEO satellite to user. $g_j(t, f)$ is the gain of the channel, where $L_{s,j}$ denotes the number of multipaths, $g_{j,l}$ represents the complex-valued gain of path l , $v_{j,l}^{\text{ut}}$ is the Doppler shift which results from the movement of user, and $\tau_{j,l}^{\text{ut}} \triangleq \tau_{j,l} - \tau_j^{\text{min}}$. We assume that the delay and the Doppler shift due to the LEO satellite is able to be compensated at each user. It is important to note that LEO satellite communication systems generally operate under line-of-sight (LoS) conditions, and the channel is thus able to be modelled using the Rician channel model [5]. $g_j(t, f)$ is assumed to follow a Rician distribution with a Rician factor of κ_m and an average power of γ , where $\gamma = \mathbb{E}[g_j(t, f)]^2$.

The response vector $\mathbf{v}_j$ of the UPA channel is defined as

$$\mathbf{v}_j(\theta_j, \varphi_j) \triangleq \mathbf{v}_x(\theta_j, \varphi_j) \otimes \mathbf{v}_y(\theta_j), \quad (3)$$

where θ_j and φ_j represent the angles relative to x-axis and y-axis, respectively. $\mathbf{v}_x(\theta_j, \varphi_j)$ and $\mathbf{v}_y(\theta_j)$ represent the UPA response vectors corresponding to the x -axis and y -axis, respectively, and are defined as follows

$$\mathbf{v}_x(\theta_j, \varphi_j) = \frac{1}{\sqrt{M_1}} [1, \exp \{-j\pi \sin(\theta_j) \cos(\varphi_j)\}, \dots, \exp \{-j\pi (M_1 - 1) \sin(\theta_j) \cos(\varphi_j)\}]^T, \quad (4)$$

$$\mathbf{v}_y(\theta_j) = \frac{1}{\sqrt{M_2}} [1, \exp \{-j\pi \cos(\theta_j)\}, \dots, \exp \{-j\pi (M_2 - 1) \cos(\theta_j)\}]^T. \quad (5)$$

In practical LEO satellite downlink scenarios, the high-speed motion of the LEO satellite often results in inaccuracies in the AoD. The actual AoD can be modelled as [8]

$$\varphi_{s,j} = \hat{\varphi}_{s,j} + \varphi_{s,j}^e + \bar{\varphi}, \quad (6a)$$

$$\theta_{s,j} = \hat{\theta}_{s,j} + \theta_{s,j}^e + \bar{\theta}, \quad (6b)$$

where $\hat{\varphi}_{s,j}$ and $\hat{\theta}_{s,j}$ denote the estimated angles of the j -th user. Angular estimation errors $\varphi_{s,j}^e$ and $\theta_{s,j}^e$ are introduced as random variables. $\bar{\varphi}$ and $\bar{\theta}$ denote the angular deviation which can be negligible [7].

Though imperfect CSI is considered for each user in the LEO satellite communication system, the distribution information of user-angle uncertainties however can be known [8]. The angle uncertainties for the j -th user, $\theta_{s,j}^e \in [-\theta_s^e, \theta_s^e]$ and $\varphi_{s,j}^e \in [-\varphi_s^e, \varphi_s^e]$, are assumed to follow a uniform distribution within the certain intervals.

B. Terrestrial Channel Model

We assume that each ground BS is equipped with a UPA of dimension $N_t = N_1 N_2$, where N_1 and N_2 correspond to the number of array elements aligned along the x and z -axes, respectively. The downlink channel between the i -th BS and the j -th user can be modelled as [15], [16]

$$\mathbf{h}_{i,j} = \sqrt{\frac{1}{L_{i,j}}} \sum_{l=1}^{L_{i,j}} \alpha_{i,j,l}^{\text{sta}} \mathbf{a}_h(\theta_{i,j,l}, \varphi_{i,j,l}) \otimes \mathbf{a}_v(\theta_{i,j,l}), \quad (7)$$

where $\text{sta} \in \{\text{los}, \text{nlos}, \text{out}\}$ denotes the path state and $\alpha_{i,j,l}$ is the complex path gain. When $\text{sta} \in \{\text{los}, \text{nlos}\}$, each path gain follows an independent and identical distribution (i.i.d.) $\mathcal{CN}(0, \eta_j)$. In contrast, the gain decays significantly when the path is in out state. $\theta_{i,j,l}$ and $\varphi_{i,j,l}$ correspond to the vertical and horizontal angles associated with the l -th path, respectively. A user is classified as a weak user when all his possible paths are all in out state. The horizontal direction vector $\mathbf{a}_h(\theta_{i,j,l}, \varphi_{i,j,l})$ and the vertical direction vector $\mathbf{a}_v(\theta_{i,j,l})$ are respectively

$$\mathbf{a}_h(\theta_{i,j,l}, \varphi_{i,j,l}) = \begin{bmatrix} e^{(-j\pi \left(\frac{N_1-1}{2}\right) \sin \theta_{i,j,l} \cos \varphi_{i,j,l})}, \dots, e^{(+j\pi \left(\frac{N_1-1}{2}\right) \sin \theta_{i,j,l} \cos \varphi_{i,j,l})} \end{bmatrix}^T, \quad (8)$$

$$\mathbf{a}_v(\theta_{i,j,l}) = \begin{bmatrix} e^{(-j\pi \left(\frac{N_2-1}{2}\right) \cos \theta_{i,j,l})}, \dots, e^{(+j\pi \left(\frac{N_2-1}{2}\right) \cos \theta_{i,j,l})} \end{bmatrix}^T. \quad (9)$$

C. Transmission Signal Model

Let $\mathbf{w}_{i,j} \in \mathbb{C}^{N_t \times 1}$ represent the precoding vector from the i -th BS to the j -th user, and $\|\mathbf{w}_{i,j}\|^2 = P_{i,j}$ be the signal transmission power. The signal $x_{b,i,j}$ transmitted by the i -th BS to the j -th user satisfies $\mathbb{E}[|x_{b,i,j}|^2] = 1$. Similarly, let $\mathbf{b}_j \in \mathbb{C}^{M_t \times 1}$ denote the satellite precoding vector for the j -th user. The signal $x_{s,j}$ sent by the satellite to j -th user also satisfies $\mathbb{E}[|x_{s,j}|^2] = 1$. Then, the received signal of the j -th user is

$$y_j = \sum_{i=1}^{N_B} \mathbf{h}_{i,j}^H \mathbf{w}_{i,j} x_{b,i,j} + \mathbf{g}_j^H \mathbf{b}_j x_{s,j} + \sum_{i=1}^{N_B} \mathbf{h}_{i,j}^H \times \sum_{\substack{j'=1 \\ j' \neq j}}^{K_u} \mathbf{w}_{i,j'} x_{b,i,j'} + \mathbf{g}_j^H \sum_{\substack{j'=1 \\ j' \neq j}}^{K_u} \mathbf{b}_{j'} x_{s,j'} + n_j, \quad (10)$$

where $n_j \sim \mathcal{CN}(0, \sigma_n^2)$ represents complex Gaussian noise with variance σ_n^2 . The noise variance is given by $\sigma_n^2 = \kappa B T$, with κ representing Boltzmann constant, B denoting the bandwidth, and T referring to the noise temperature.

D. Problem Formulation

In (10), the first two terms represent the signals, while the third and fourth terms correspond to the interference. Then, the signal-to-interference-plus-noise-ratio (SINR) of the j -th user's received signal is as follows

$$\text{SINR}_j = \frac{|\sum_{i=1}^{N_B} \mathbf{h}_{i,j}^H \mathbf{w}_{i,j} + \mathbf{g}_j^H \mathbf{b}_j|^2}{\sum_{\substack{j'=1 \\ j' \neq j}}^{K_u} |\sum_{i=1}^{N_B} \mathbf{h}_{i,j'}^H \mathbf{w}_{i,j'} + \mathbf{g}_j^H \mathbf{b}_{j'}|^2 + \sigma_n^2}. \quad (11)$$

Considering the angular estimation error that exists in the satellite downlink channel, the ergodic rate is used here to define the achievable rate of the j -th user as follows

$$R_j \triangleq \mathbb{E} \left\{ \log_2 \left(1 + \left| \sum_{i=1}^{N_B} \mathbf{h}_{i,j}^H \mathbf{w}_{i,j} + \mathbf{g}_j^H \mathbf{b}_j \right|^2 \right) \right. \\ \left. \mathbb{E} \left\{ \sum_{\substack{j'=1 \\ j' \neq j}}^{K_u} \left| \sum_{i=1}^{N_B} \mathbf{h}_{i,j'}^H \mathbf{w}_{i,j'} + \mathbf{g}_j^H \mathbf{b}_{j'} \right|^2 + \sigma_n^2 \right\} \right\}. \quad (12)$$

The objective is to maximize the sum rate of the cooperative STN through precoding design. We set power thresholds for both the LEO satellite and BSs. Therefore, the entire optimization problem can be formulated as follows

$$\mathcal{P}_1 : \max_{\mathbf{w}_{i,j}, \mathbf{b}_j} \sum_{j=1}^{K_u} R_j, \quad (13a)$$

$$\text{s.t.} \quad \sum_{j=1}^{K_u} \|\mathbf{w}_{i,j}\|^2 \leq P_{B,\max}, i \in [1, N_B], \quad (13b)$$

$$\sum_{j=1}^{K_u} \|\mathbf{b}_j\|^2 \leq P_{S,\max}. \quad (13c)$$

III. ROBUST PRECODING DESIGN

The objective function of optimization problem $\mathcal{P}_1$ is non-convex, making it difficult to obtain an exact solution. In this section, we apply certain transformations to address the problem.

A. Rate Approximation and Semidefinite Relaxation (SDR)

For simplicity, we let $\mathbf{w}_j = [\mathbf{b}_j^H, \mathbf{w}_{1,j}^H, \dots, \mathbf{w}_{N_B,j}^H]^H$, $\mathbf{h}_j = [\mathbf{g}_j^H, \mathbf{h}_{1,j}^H, \dots, \mathbf{h}_{N_B,j}^H]^H$. Thus, $\|\mathbf{w}_j\|^2 = \text{Tr}(\mathbf{w}_j \mathbf{w}_j^H)$ and $\|\mathbf{w}_j^H \mathbf{h}_j\|^2 = \text{Tr}(\mathbf{w}_j \mathbf{w}_j^H \mathbf{h}_j \mathbf{h}_j^H)$. Then, (13b) and (13c) can be modified to

$$\sum_{j=1}^{K_u} \text{Tr}(A_{B,i} \mathbf{w}_j \mathbf{w}_j^H) \leq P_{B,\max}, i \in [1, N_B], \quad (14a)$$

$$\sum_{j=1}^{K_u} \text{Tr}(A_S \mathbf{w}_j \mathbf{w}_j^H) \leq P_{S,\max}, \quad (14b)$$

where $A_{B,i}$ and A_S are diagonal matrices. The elements on the diagonal corresponding to the satellite or any one of BSs are 1, and the remaining elements are 0. We introduce the optimization variables as $\{\mathbf{W}_j \triangleq \mathbf{w}_j \mathbf{w}_j^H\}_{j=1}^{K_u}$. Here, $\mathbf{W}_j$ is subject to the rank constraint that $\text{rank}(\mathbf{W}_j) = 1$. Additionally, we define $\{\mathbf{H}_j \triangleq \mathbf{h}_j \mathbf{h}_j^H\}_{j=1}^{K_u}$. To simplify the computation of R_j , we utilize the $\bar{R}_j$ in [17], [18] as below

$$R_j \leq \bar{R}_j \triangleq \log_2 \left(1 + \frac{\mathbb{E}\{\text{Tr}(\mathbf{H}_j \mathbf{W}_j)\}}{\mathbb{E}\{\sum_{j' \neq j} \text{Tr}(\mathbf{H}_j \mathbf{W}_{j'}) + \sigma_n^2\}} \right) \quad (15a)$$

$$= F_j(\mathbf{W}) - G_j(\mathbf{W}), \quad (15b)$$

where

$$\bar{\mathbf{H}}_j = \mathbb{E}\{\mathbf{H}_j\} = \mathbb{E}\{\mathbf{h}_j \mathbf{h}_j^H\}, \quad (16a)$$

$$F_j(\mathbf{W}) = \log_2(\text{Tr}(\bar{\mathbf{H}}_j \sum_{j' \neq j} \mathbf{W}_{j'}) + \sigma_n^2), \quad (16b)$$

$$G_j(\mathbf{W}) = \log_2(\text{Tr}(\bar{\mathbf{H}}_j \sum_{j' \neq j} \mathbf{W}_{j'}) + \sigma_n^2). \quad (16c)$$

Note that matrix $\bar{\mathbf{H}}_j$ serves to mitigate the impact of channel angle uncertainties. The derivation of $\bar{\mathbf{H}}_j$ is presented below.

B. Derivation of $\bar{\mathbf{H}}_j$

As can be observed from the structure of $\mathbf{h}_j$

$$[\underbrace{\dots}_{\text{Satellite}} \underbrace{\dots}_{\text{BS}_1} \underbrace{\dots}_{\text{BS}_2} \dots \underbrace{\dots}_{\text{BS}_{N_B}}],$$

the first M_t elements are generated by the satellite downlink channel, and every N_t elements thereafter are generated by a BS downlink channel. We denote m as the index of the element in the joint channel vector $\mathbf{h}_j$. When $m \leq M_t$, the element is generated by the satellite downlink channel. From the above discussion related to LEO satellite downlink channel, the m -th element of $\mathbf{v}_j$ is

$$(\mathbf{v}_j)_m = \frac{1}{\sqrt{M_t}} e^{-j\pi t_1 \sin(\hat{\theta}_{s,j} + \theta_{s,j}^e) \cos(\hat{\varphi}_{s,j} + \varphi_{s,j}^e)} \times e^{-j\pi(m-t_1 M_2 - 1) \cos(\hat{\theta}_{s,j} + \theta_{s,j}^e)},$$

$$m = 1, \dots, M_t, t_1 = \lceil \frac{m}{M_2} \rceil - 1, \quad (17)$$

the m -th element of $\mathbf{h}_j$ now is therefore $e^{j2\pi[t v_j^{\text{sat}} - f \tau_j^{\text{min}}]} \cdot g_j(t, f) \cdot (\mathbf{v}_j)_m$. After performing Doppler compensation on the received signal at each user, this can be concisely represented as $g_j(\mathbf{v}_j)_m$.

When $M_t < m$, we denote $i_m = \lceil \frac{m-M_t}{N_t} \rceil$ as the label of one BS and k as the index of the element in the channel vector part of this BS. As a result, k is expressed as

$$k = f(m) = m - M_t - \left(\left\lceil \frac{m - M_t}{N_t} \right\rceil - 1 \right) N_t. \quad (18)$$

According to the above discussion on the BS downlink channel, It is evident that the k -th element of the i_m -th BS channel vector segment is

$$\mathbf{h}_{i_m,j,k} = \sqrt{\frac{1}{L_{i_m,j}}} \sum_{l=1}^{L_{i_m,j}} \alpha_{i_m,j,l}^{\text{sta}} \exp \left(-j\frac{\pi}{2} (N_1 + 1 - 2t_2) \right. \\ \left. \times \sin \theta_{i_m,j,l} \cos \varphi_{i_m,j,l} - j\frac{\pi}{2} [1 + (2t_2 - 1)N_2 - 2k] \cos \theta_{i_m,j,l} \right), t_2 = \left\lceil \frac{k}{N_2} \right\rceil. \quad (19)$$

Thus, the (m, n) -th element in $\mathbf{H}_j$ is given as follows

$$[\mathbf{H}_j = \mathbf{h}_j \mathbf{h}_j^H]_{m,n} = \begin{cases} |g_j|^2 (\mathbf{v}_j)_m (\mathbf{v}_j)_n^H, & m \leq M_t, n \leq M_t, \\ g_j (\mathbf{v}_j)_m (\mathbf{h}_{i_m,j})_{f(n)}^H, & m \leq M_t, n > M_t, \\ g_j^H (\mathbf{v}_j)_n^H (\mathbf{h}_{i_m,j})_{f(m)}, & m > M_t, n \leq M_t, \\ (\mathbf{h}_{i_m,j})_{f(m)} (\mathbf{h}_{i_m,j})_{f(n)}^H, & m > M_t, n > M_t. \end{cases} \quad (20)$$

In (15b), we use $\bar{\mathbf{H}}_j$ not $\mathbf{H}_j$ to obtain the objective function. As a result, in the following, we need to get the real expression of $\bar{\mathbf{H}}_j$ in each case.

For $m \leq M_t, n \leq M_t$, the result is as follows [8]

$$\mathbb{E} \left\{ |g_j|^2 (\mathbf{v}_j)_m (\mathbf{v}_j)_n^H \right\} = \begin{cases} \frac{\gamma}{M_1 M_2}, & n = m, t_1 = t_3; \\ \frac{\gamma}{M_1 M_2 \theta_s^e} \frac{\exp\{j\pi(n-m) \cos(\hat{\theta}_{s,j})\} \sin(\pi(n-m) \sin(\hat{\theta}_{s,j}) \theta_s^e)}{\pi(n-m) \sin(\hat{\theta}_{s,j})}, & n \neq m, t_1 = t_3; \\ \frac{\gamma}{M_1 M_2} \frac{Z}{\theta_s^e \varphi_s^e} \frac{\sin(D \varphi_s^e \sin(\hat{\theta}_{s,j})) \sin(\pi C D \theta_s^e)}{\pi C D \sin(\hat{\theta}_{s,j})}, & n \neq m, t_1 \neq t_3, \end{cases} \quad (21)$$

where

$$\begin{aligned}
C &= (t_3 - t_1) \cos(\hat{\varphi}_{s,j}) \cos(\hat{\theta}_{s,j}) \\
&\quad - (n - m - (t_3 - t_1)M_1) \sin(\hat{\theta}_{s,j}), \\
D &= \pi(t_3 - t_1) \sin(\hat{\varphi}_{s,j}), \\
Z &= \exp \left\{ \bar{J}\pi(n - m - (t_3 - t_1)M_2) \cos(\hat{\theta}_{s,j}) \right. \\
&\quad \left. + \bar{J}\pi(t_3 - t_1) \cos(\hat{\varphi}_{s,j}) \sin(\hat{\theta}_{s,j}) \right\}, \\
t_1 &= \left\lfloor \frac{m}{M_2} \right\rfloor - 1, t_3 = \left\lfloor \frac{n}{M_2} \right\rfloor - 1.
\end{aligned} \tag{22}$$

For $m \leq M_t, n > M_t$,

$$\begin{aligned}
\mathbb{E} \left\{ g_j(\mathbf{v}_j)_m (\mathbf{h}_{i_n,j})_{f(n)}^H \right\} &= \mathbb{E} \{ g_j \} \cdot \mathbb{E} \{ (\mathbf{v}_j)_m \} \cdot (\mathbf{h}_{i_n,j})_{f(n)}^H \\
&= \frac{\gamma_m}{\sqrt{M_t}} \mathbb{E} \left\{ \exp \left(-\bar{J}\pi t_1 \sin(\hat{\theta}_{s,j} + \theta_{s,j}^e) \cos(\hat{\varphi}_{s,j} + \varphi_{s,j}^e) \right. \right. \\
&\quad \left. \left. - \bar{J}\pi(m - t_1 M_2 - 1) \cos(\hat{\theta}_{s,j} + \theta_{s,j}^e) \right) \right\} \cdot (\mathbf{h}_{i_n,j})_{f(n)}^H,
\end{aligned} \tag{23}$$

where γ_m is the mean value of $g_j(t, f)$. We employ the known information about the error distribution to get the result. $\varphi_{s,j}^e$ is first integrated, and all other random variables are treated as constants. We first consider integral $\int_{-\varphi_s^e}^{\varphi_s^e} \frac{1}{2\varphi_s^e} e^{-\bar{J}\pi t_1 \sin(\hat{\theta}_{s,j} + \theta_{s,j}^e) \cos(\hat{\varphi}_{s,j} + \varphi_{s,j}^e)} d\varphi_{s,j}^e$. However, the integral is not integrable. It is worth noting that $\varphi_{s,j}^e$ and other random variables vary around 0 and have smaller values. Thus, $\cos(\varphi_{s,j}^e) \approx 1$ and $\sin(\varphi_{s,j}^e) \approx \varphi_{s,j}^e$ are reasonable. As a result, we make the approximations below

$$\cos(\hat{\varphi}_{s,j} + \varphi_{s,j}^e) \approx \cos(\hat{\varphi}_{s,j}) - \sin(\hat{\varphi}_{s,j})\varphi_{s,j}^e, \tag{24}$$

$$\sin(\hat{\theta}_{s,j} + \theta_{s,j}^e) \approx \sin(\hat{\theta}_{s,j}) + \cos(\hat{\theta}_{s,j})\theta_{s,j}^e, \tag{25}$$

$$\cos(\hat{\theta}_{s,j} + \theta_{s,j}^e) \approx \cos(\hat{\theta}_{s,j}) - \sin(\hat{\theta}_{s,j})\theta_{s,j}^e. \tag{26}$$

As a result,

$$\begin{aligned}
&\int_{-\varphi_s^e}^{\varphi_s^e} \frac{1}{2\varphi_s^e} e^{-\bar{J}\pi t_1 \sin(\hat{\theta}_{s,j} + \theta_{s,j}^e) \cos(\hat{\varphi}_{s,j} + \varphi_{s,j}^e)} d\varphi_{s,j}^e \\
&= \begin{cases} \frac{e^{-\bar{J}\pi t_1 \sin(\hat{\theta}_{s,j} + \theta_{s,j}^e) \cos(\hat{\varphi}_{s,j})} \sin(\pi t_1 \sin(\hat{\theta}_{s,j} + \theta_{s,j}^e) \sin(\hat{\varphi}_{s,j}) \varphi_s^e)}{\varphi_s^e \pi t_1 \sin(\hat{\theta}_{s,j} + \theta_{s,j}^e) \sin(\hat{\varphi}_{s,j})}, & t_1 \neq 0; \\ 1, & t_1 = 0. \end{cases}
\end{aligned} \tag{27}$$

If $t_1 = 0, m \neq 1$, then we integrate $\theta_{s,j}^e$ as follows

$$\begin{aligned}
&\int_{-\theta_s^e}^{\theta_s^e} \frac{1}{2\theta_s^e} e^{-\bar{J}\pi(m-1) \cos(\hat{\theta}_{s,j} + \theta_{s,j}^e) \sin(\hat{\varphi}_{s,j}) \varphi_s^e} d\theta_{s,j}^e \\
&= \frac{e^{-\bar{J}\pi(m-1) \cos(\hat{\theta}_{s,j}) \sin(\hat{\varphi}_{s,j}) \varphi_s^e}}{(m-1)\pi \sin(\hat{\theta}_{s,j}) \theta_s^e} \sin[(m-1)\pi \sin(\hat{\theta}_{s,j}) \theta_s^e].
\end{aligned} \tag{28}$$

While if $t_1 = 0$ and $m = 1$, the result is 1.

We continue with the case of $t_1 \neq 0$. According to the formula (27), $\theta_{s,j}^e$ cannot be directly integrated. Let $A_0 = \pi t_1 \sin(\hat{\varphi}_{s,j}) \varphi_s^e$, there are

$$\sin(A_0 \sin(\hat{\theta}_{s,j} + \theta_{s,j}^e)) \approx \sin[A_0 \sin(\hat{\theta}_{s,j})]. \tag{29}$$

In the same way, $A_0[\sin(\hat{\theta}_{s,j}) + \cos(\hat{\theta}_{s,j})\theta_{s,j}^e] \approx A_0 \sin(\hat{\theta}_{s,j})$. We make $A_1 = A_0 \sin(\hat{\theta}_{s,j})$. Then, the integral part about $\theta_{s,j}^e$ is

$$\begin{aligned}
&\int_{-\theta_s^e}^{\theta_s^e} \frac{1}{2\theta_s^e} \frac{\sin(A_1)}{A_1} \exp(-\bar{J}\pi t_1 \cos(\hat{\varphi}_{s,j})[\sin(\hat{\theta}_{s,j}) \\
&\quad + \cos(\hat{\theta}_{s,j})\theta_{s,j}^e] - \bar{J}\pi(m - t_1 M_2 - 1)[\cos(\hat{\theta}_{s,j}) \\
&\quad - \sin(\hat{\theta}_{s,j})\theta_{s,j}^e]) d\theta_{s,j}^e \\
&= \begin{cases} \frac{B_1}{B_2} \sin(B_2), & B_2 \neq 0, \\ B_1, & B_2 = 0, \end{cases}
\end{aligned} \tag{30}$$

where

$$B_1 = \frac{\sin(A_1)}{A_1} e^{-\bar{J}\pi t_1 \cos \hat{\varphi}_{s,j} \sin(\hat{\theta}_{s,j}) - \bar{J}\pi(m - t_1 M_2 - 1) \cos(\hat{\theta}_{s,j})}, \tag{31}$$

$$B_2 = \pi \theta_s^e [(m - t_1 M_2 - 1) \sin(\hat{\theta}_{s,j}) - t_1 \cos(\hat{\varphi}_{s,j}) \cos(\hat{\theta}_{s,j})]. \tag{32}$$

The third case is analogous to the second case and requires no further discussion. As for the fourth case, since it does not involve a random variable, its mean is equal to itself. At this point, we complete the derivation of $\bar{\mathbf{H}}_j$.

C. Concave-Convex Procedure (CCCP)

The objective function in (15b) remains non-convex but conforms to the structure of a difference of concave functions. Here it is converted to a convex function using the CCCP method [19]. To be specific, we handle the second term in (15b) with a first-order Taylor series expansion and rewrite the problem $\mathcal{P}_1$ as follows

$$\mathcal{P}_2 : \max_{\{\mathbf{W}_j\}_1^{K_u}} \sum_{j=1}^{K_u} F_j(\mathbf{W}) - \Delta G_j^{(\tau)}(\mathbf{W}), \tag{33a}$$

$$\text{s.t. (14a), (14b)}, \tag{33b}$$

$$\mathbf{W}_j \succeq 0, \forall j \in [1, K_u], \tag{33c}$$

where

$$\begin{aligned}
\Delta G_j^{(\tau)}(\mathbf{W}) &= G_j(\mathbf{W}^{(\tau)}) + \sum_{j' \neq j} \text{Tr} \left\{ \left(\frac{\partial G_j(\mathbf{W}^{(\tau)})}{\partial \mathbf{W}_{j'}} \right)^T \right. \\
&\quad \left. \times (\mathbf{W}_{j'} - \mathbf{W}_{j'}^{(\tau)}) \right\},
\end{aligned} \tag{34a}$$

$$\frac{\partial G_j(\mathbf{W}^{(\tau)})}{\partial \mathbf{W}_{j'}} = \frac{\bar{\mathbf{H}}_j^T}{\left(\text{Tr}(\bar{\mathbf{H}}_j \sum_{\ell \neq j} \mathbf{W}_\ell^{(\tau)}) + \sigma_n^2 \right) \ln 2}. \tag{34b}$$

$\mathbf{W}_j^{(\tau)}$ is the set of variables $\{\mathbf{W}_1^{(\tau)}, \dots, \mathbf{W}_{K_u}^{(\tau)}\}$, with τ being the iteration index. The problem $\mathcal{P}_2$, being convex in nature, can be efficiently addressed using convex optimization techniques [17]. Typically, the resulting solutions adhere to the rank constraint, i.e., $\text{rank}(\mathbf{W}_j) = 1$. The final beamformers can be derived through eigenvalue decomposition or Gaussian randomization method if the rank constraint is not satisfied [20]. The complexity of our proposed algorithm is approximately $\mathcal{O}(L_c K_u (M_t + N_B \cdot N_t)^3)$, where L_c denotes the number of iterations in the CCCP algorithm.

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Orbit altitude	$H = 600$ km
Carrier frequency	$f = 2$ GHz
Equivalent noise temperature	$T = 300$ K
Number of users	$K_u = 4$
Bandwidth	$B = 20$ MHz
Satellite transmitter gain	$G_t^{\text{leo}} = 50$ dBi
BS transmitter gain	$G_t^{\text{bs}} = 18$ dBi
Receiver gain	$G_r = 0$ dB
Satellite antenna	8×8
BS antenna	4×4
Rice factor	$\kappa_m = 5$ dB
Satellite maximum transmit power	$P_{S,\max} = 20$ W
Boltzmann constant	$\kappa = 1.38 \times 10^{-23}$ Joule/K

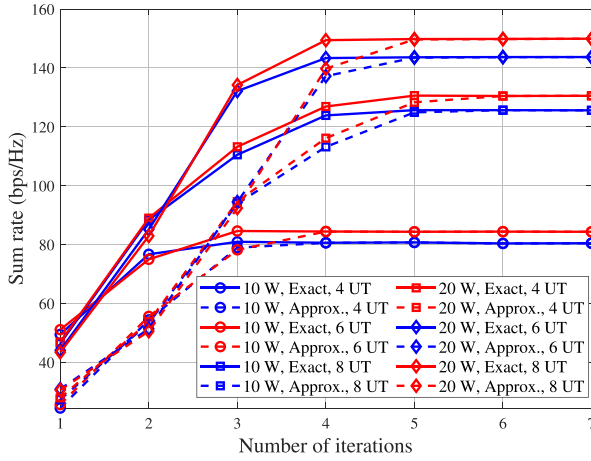


Fig. 1. The convergence performance of our proposed approach.

IV. SIMULATIONS RESULTS

This section presents an evaluation of the performance of our proposed precoding scheme. For simplicity, the number of BSs in simulation is set to 1. One user was in outage state and the channel gain was attenuated by 20 dB. We call this user as the weak user. γ is calculated as $\gamma = G_t^{\text{leo}}(c/(4\pi f_c H))^2$. η_j is calculated as $\eta_j = G_t^{\text{bs}}(c/(4\pi f_c D_j))^2$, where D_j denotes the distance from BS to j -the user. The AoD error ranges are $\theta_s^e = \varphi_s^e = 0.1^\circ$. Table I presents the key simulation parameters and their corresponding values.

Fig. 1 illustrates the convergence performance of our proposed algorithm. The power shown in the legend is the BS maximum transmit power. The simulation results indicate that the algorithm converges smoothly within 4 to 6 iterations. The approximate case means computing the sum rate by (33a). The error between the exact and the approximate sum rate is small enough to ignore when the algorithm converges. As the number of users increases, the growth rate of the system sum-rate performance diminishes and eventually converges due to the power constraints of both the satellite and the BS.

Fig. 2 compares the rate performance of weak users in the cooperative STN scenario and the BS-only scenario. The dB values in the legend represent the additional loss coefficient of the channel between the BS and the weak user. As the loss increases, the achievable rate of the weak user in the BS-only scenario degrades sharply, whereas the cooperative STN scenario exhibits a more gradual decline, leading to a

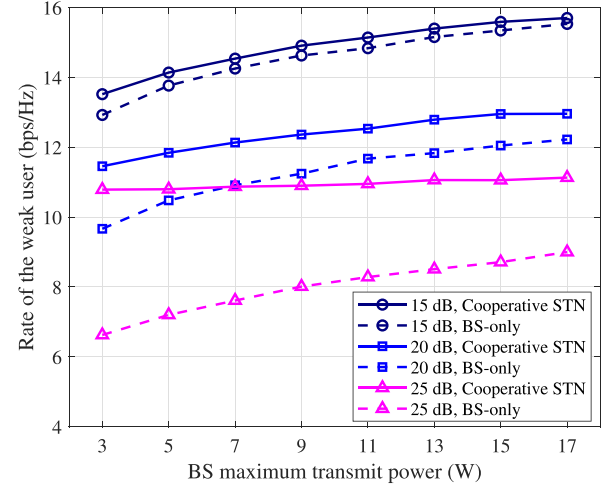


Fig. 2. The rate performance of the weak user.

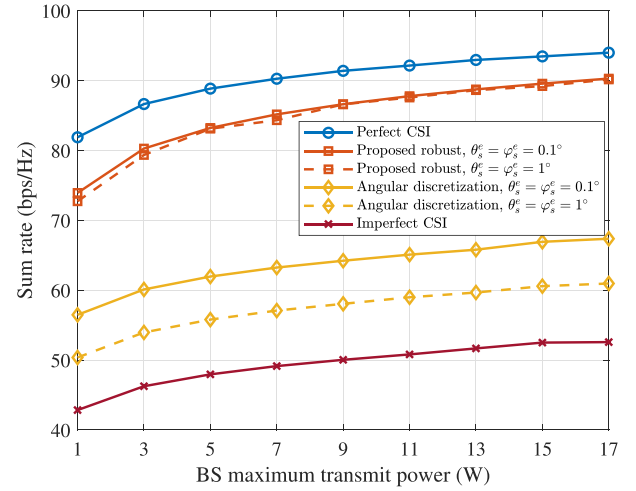


Fig. 3. The sum rate performance of different algorithms.

widening performance gap between the two schemes. Therefore, when the channel between the BS and the users is prone to severe attenuation due to obstacles or other impairments, cooperative STN can effectively enhance communication reliability.

An angular discretization method in [12] is adopted for comparative evaluation with our proposed scheme. This method first determines the upper and lower bounds of the angular uncertainty region, then uniformly samples discrete angle points within the interval, with each point corresponding to a deterministic channel vector. The actual channel is approximated via a weighted sum of these discrete vectors. This approach is grounded in convex hull theory, which guarantees that any channel vector within the uncertainty region can be represented as a convex combination of discrete channel vectors. We uniformly sample 500 discrete points within each uncertainty region and assign equal weights to all channel vectors [12].

Fig. 3 depicts the system sum rate performance of different algorithms as a function of the BS maximum transmit power. Here, each user's AoD errors are set to $\theta_{s,j}^e = \varphi_{s,j}^e = 0.1^\circ$. The AoD error ranges are used as pre-known information in our proposed robust and angular discretization algorithms for robust precoding. In contrast, the imperfect case directly employs the estimated channel for precoding. It is shown that the performance of our robust precoding algorithm

closely matches that achieved with perfect CSI for precoding. The imperfect case exhibits significant performance degradation compared to others, further highlighting the effectiveness of our precoding algorithm. With the increase of AoD error ranges, the performance of the angular discretization algorithm decreases significantly. However, by fully exploiting the statistical distribution characteristics of AoD errors, the proposed robust algorithm is less affected by the increase of AoD error ranges and demonstrates significantly superior performance compared to the conventional angular discretization algorithm.

V. CONCLUSION

In conclusion, this paper addressed the robust precoding scheme in cooperative massive MIMO LEO STNs. We began by formulating the optimization problem for downlink ergodic sum rate maximization under energy constraints. The rate function was closely approximated through a tight closed-form expression. Then, we relaxed the rank constraint through the SDR procedure, and introduced intermediate variables to mitigate the impact of imperfect CSI, leveraging a priori knowledge of the channel angular error distribution. Given the non-convex nature of both the objective and the constraints, the CCCP method was utilized to address the problem. The proposed robust precoding scheme was shown to be effective through simulation results.

ACKNOWLEDGMENT

The authors would like to thank the editor and the anonymous reviewers for their constructive comments and suggestions.

REFERENCES

- [1] S. Chen, Y.-C. Liang, S. Sun, S. Kang, W. Cheng, and M. Peng, "Vision, requirements, and technology trend of 6G: How to tackle the challenges of system coverage, capacity, user data-rate and movement speed," *IEEE Wireless Commun.*, vol. 27, no. 2, pp. 218–228, Apr. 2020.
- [2] J. G. Andrews, T. E. Humphreys, and T. Ji, "6G takes shape," *IEEE BITS Inf. Theory Mag.*, vol. 4, no. 1, pp. 2–24, Mar. 2024.
- [3] L. You et al., "Ubiquitous integrated sensing and communications for massive MIMO LEO satellite systems," *IEEE Internet Things Mag.*, vol. 7, no. 4, pp. 30–35, Jul. 2024.
- [4] 3rd Generation Partnership Project (3GPP), "Study on new radio (NR) to support non-terrestrial networks (release 15)," *3GPP, Sophia Antipolis, France*, Tech. Specification (TS) 38.811, v15.4.0, Sep. 2020.
- [5] L. You, K.-X. Li, J. Wang, X. Gao, X.-G. Xia, and B. Ottersten, "Massive MIMO transmission for LEO satellite communications," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 8, pp. 1851–1865, Aug. 2020.
- [6] Y. Huang, L. You, C. G. Tsinos, W. Wang, and X. Gao, "QoS-aware precoding in downlink massive MIMO LEO satellite communications," *IEEE Commun. Lett.*, vol. 27, no. 6, pp. 1560–1564, Jun. 2023.
- [7] D. Cilden-Guler, M. Raitoharju, R. Piche, and C. Hajiyev, "Nanosatellite attitude estimation using Kalman-type filters with non-Gaussian noise," *Aerosp. Sci. Technol.*, vol. 92, pp. 66–76, Sep. 2019.
- [8] Y. Liu, Y. Wang, J. Wang, L. You, W. Wang, and X. Gao, "Robust downlink precoding for LEO satellite systems with per-antenna power constraints," *IEEE Trans. Veh. Technol.*, vol. 71, no. 10, pp. 10694–10711, Oct. 2022.
- [9] Z. Lin, M. Lin, J.-B. Wang, Y. Huang, and W.-P. Zhu, "Robust secure beamforming for 5G cellular networks coexisting with satellite networks," *IEEE J. Sel. Areas Commun.*, vol. 36, no. 4, pp. 932–945, Apr. 2018.
- [10] D. Peng, A. Bandi, Y. Li, S. Chatzinotas, and B. Ottersten, "Hybrid beamforming, user scheduling, and resource allocation for integrated terrestrial-satellite communication," *IEEE Trans. Veh. Technol.*, vol. 70, no. 9, pp. 8868–8882, Sep. 2021.
- [11] Y. Zhang, L. Yin, C. Jiang, and Y. Qian, "Joint beamforming design and resource allocation for terrestrial-satellite cooperation system," *IEEE Trans. Commun.*, vol. 68, no. 2, pp. 778–791, Feb. 2020.
- [12] B. Zhao, M. Lin, M. Cheng, J.-B. Wang, J. Cheng, and M.-S. Alouini, "Robust downlink transmission design in IRS-assisted cognitive satellite and terrestrial networks," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 8, pp. 2514–2529, Aug. 2023.
- [13] J. Chu and X. Chen, "Robust design for integrated satellite-terrestrial Internet of Things," *IEEE Internet Things J.*, vol. 8, no. 11, pp. 9072–9083, Jun. 2021.
- [14] H. Zhang et al., "Robust resource allocation for integrated satellite-terrestrial communication systems with eavesdroppers," *IEEE Internet Things J.*, early access, Feb. 13, 2025, doi: [10.1109/IIOT.2025.3535410](https://doi.org/10.1109/IIOT.2025.3535410).
- [15] Z. Li, S. Yang, and T. Clessienne, "A general rate splitting scheme for hybrid precoding in mmWave systems," in *Proc. IEEE Int. Conf. Commun.*, May 2019, pp. 1–6.
- [16] L. You et al., "Network massive MIMO transmission over millimeter-wave and terahertz bands: Mobility enhancement and blockage mitigation," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 12, pp. 2946–2960, Dec. 2020.
- [17] X. Xiao, L. You, J. Wang, W. Wang, and X. Gao, "Multigroup multicast beamforming for high throughput GEO satellite communications under power-consumption outage constraints," *IEEE Commun. Lett.*, vol. 27, no. 3, pp. 941–945, Mar. 2023.
- [18] L. You et al., "Hybrid analog/digital precoding for downlink massive MIMO LEO satellite communications," *IEEE Trans. Wireless Commun.*, vol. 21, no. 8, pp. 5962–5976, Aug. 2022.
- [19] B. Jiang, Y. Yan, L. You, J. Wang, W. Wang, and X. Gao, "Robust secure transmission for satellite communications," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 2, pp. 1598–1612, Apr. 2023.
- [20] W. Wang et al., "Resource efficiency optimization for robust beamforming in multi-beam satellite communications," *IEEE Trans. Veh. Technol.*, vol. 70, no. 7, pp. 6958–6968, Jul. 2021.